

WEEKLY PRESENTATION

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NEW METHODS

Sparse Jacobian Class

```
std::size_t row_or_col_num_colors(ptr_arr, idx_arr)
```

```
    build_cig_from_compressed(ptr_arr, idx_arr, n_or_m)
```

```
    color_graph(graph, max_color)
```

```
std::size_t max_nnz_row_or_col(ptr_arr)
```

TESTS

CSR and CSC representation. Test with Eigen

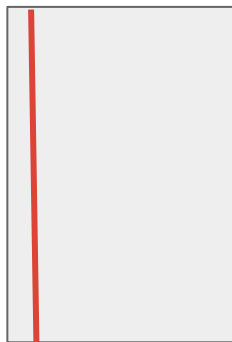
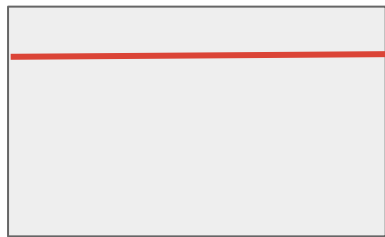
Sparse Multiplication. Tests with Eigen.

Graph and Graph coloring. gdb.

NEW FORMULATION

Matrix-matrix multiplication cost:

$$\text{nnz_product} * \min(\text{max_row_or_col})$$

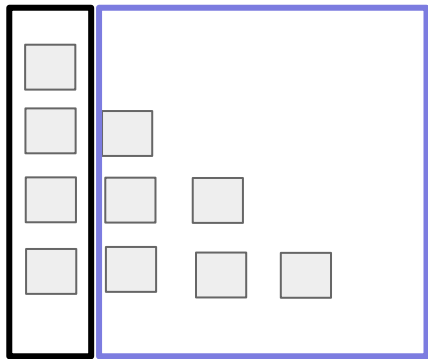


NEW FORMULATION

Motivation:

$$\begin{array}{ccc} x & \underline{x} & 0 \\ x & \underline{x} & 0 \\ 0 & 0 & x \end{array} \quad \begin{array}{ccc} 0 & 0 & x \\ 0 & 0 & x \\ x & \underline{x} & 0 \end{array} = \begin{array}{ccc} 0 & 0 & x \\ 0 & 0 & x \\ x & \underline{x} & 0 \end{array}$$

FUTURE WORK



Points to
Jacobian
Chain

Owens a
Jacobian
object

Rewrite `behaviour_fill_table.hpp`

Implement Sparse Jacobian DP

Test it.

Analyse results.

Generalize the Generator class
method to include the case shown
as motivation for the new
formulation.